Q29536 · KPYR_CANLF

Protein
Pyruvate kinase PKLR

Gene
PKLR

Status
UniProtKB reviewed (Swiss-Prot)

Organism
Canis lupus familiaris (Dog) (Canis familiaris)

Amino acids
574 (go to sequence)

Protein existence
Evidence at transcript level

Annotation score
4/5

Function

Pyruvate kinase that catalyzes the conversion of phosphoenolpyruvate to pyruvate with the synthesis of ATP, and which plays a key role in glycolysis. Also produces the side product 2-phospholactate which can inhibit fructose-2,6-bisphosphate production. 2-phospholactate can be dephosphorylated by PGP which prevents the inhibition of fructose-2,6-bisphosphate production and allows glycolysis to occur. By Similarity

Miscellaneous

There are 4 isozymes of pyruvate kinase in mammals: L, R, M1 and M2. L type is major isozyme in the liver, R is found in red cells, M1 is the main form in muscle, heart and brain, and M2 is found in early fetal tissues. Curated

Catalytic activity

Rhea:18157

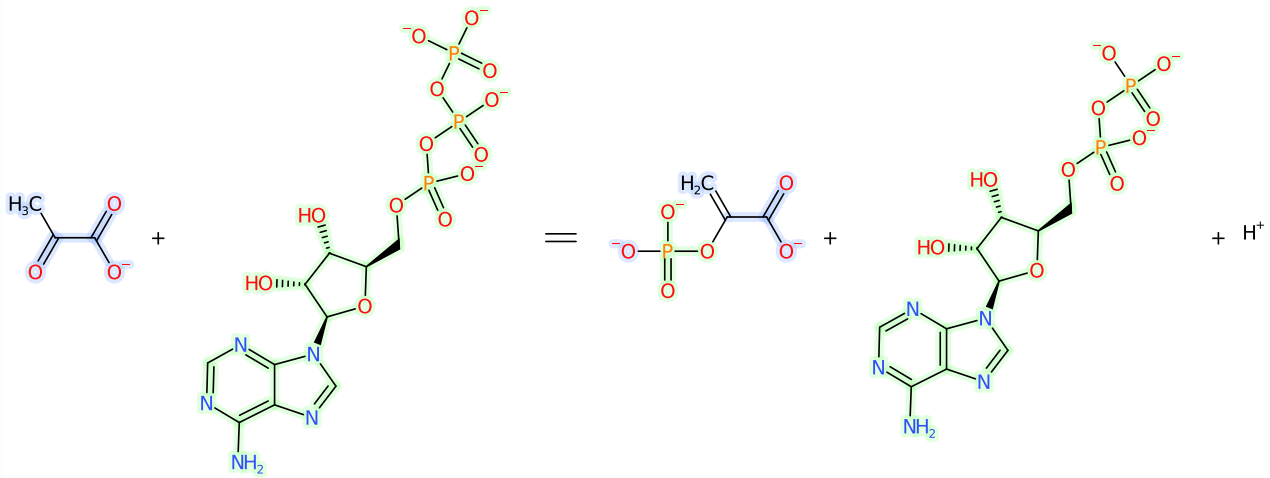
pyruvate + ATP = phosphoenolpyruvate + ADP + H^+ By Similarity
This reaction proceeds in the backward direction. By Similarity
EC:2.7.1.40 (UniProtKB | ENZYME | Rhea)

Hide reaction details

Reaction Atom map

This website requires cookies, and the limited processing of your personal data in order to function. By using the site you are agreeing to this as outlined in our **Privacy Notice**. I agree, dismiss this banner

Color atoms ☑, Number atoms ☐, Confidence = 0.34795



pyruvate » ATP » phosphoenolpyruvate » ADP » H^+ »

Rhea:86567

(S)-lactate + ATP = (2S)-2-phospholactate + ADP + H^+ By Similarity

View reaction details

Cofactor[i]

Protein has several cofactor binding sites:

Mg^{2+} (UniProtKB | Rhea | CHEBI:18420) By Similarity
Mn^{2+} (UniProtKB | Rhea | CHEBI:29035) By Similarity

K^+ (UniProtKB | Rhea | CHEBI:29103) By Similarity

Activity regulation[i]

Allosterically activated by fructose 1,6-bisphosphate. By Similarity

This website requires cookies, and the limited processing of your personal data in order to function. By using the site you are agreeing to this as outlined in our **Privacy Notice**.

Features

Showing features for binding site[i], site[i].

±	TYPE	ID	POSITION(S)	DESCRIPTION	
+	Binding site		116	substrate By Similarity	
+	Binding site		118	K^+ (UniProtKB | ChEBI) By Similarity	
+	Binding site		118-121	ATP (UniProtKB | ChEBI) By Similarity	Tools Add
+	Binding site		120	K^+ (UniProtKB | ChEBI) By Similarity	
+	Binding site		156	K^+ (UniProtKB | ChEBI) By Similarity	
+	Binding site		157	K^+ (UniProtKB | ChEBI) By Similarity	

Expand table

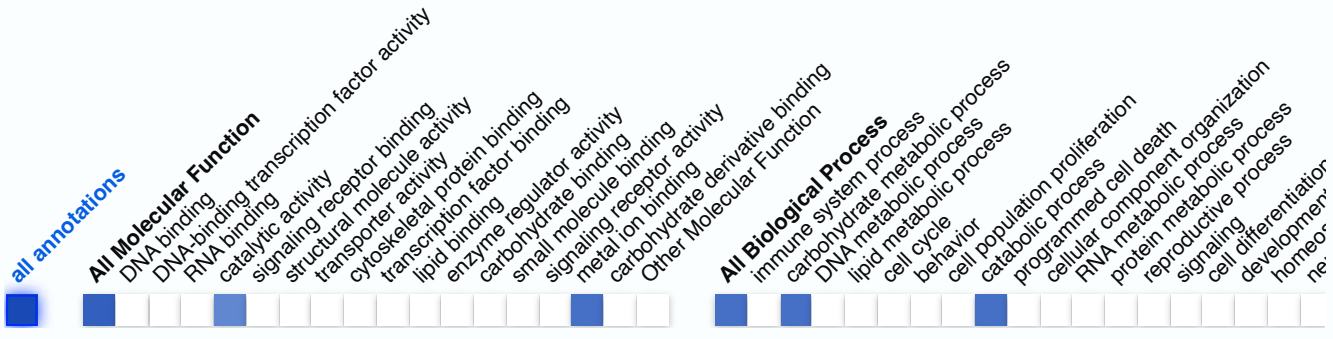
Gene Ontology[i]

GO annotations | GO-CAM models New

Gene Ontology (GO) annotations organized by slimming set.

Slimming set:

Animals and fungi (Opisthokonta)



Cell color indicative of number of GO terms

This website requires cookies, and the limited processing of your personal data in order to function. By using the site you are agreeing to this as outlined in our **Privacy Notice**.

ASPECT	TERM
Molecular Function	potassium ion binding Source:InterPro
Molecular Function	pyruvate kinase activity Source:UniProtKB
Biological Process	cellular response to insulin stimulus Source:GO_Central
Biological Process	glycolytic process Source:GO_Central

Access the complete set of GO annotations on QuickGO

Keywords

Molecular function

#Allosteric enzyme
#Kinase
#Transferase

Biological process

#Glycolysis

Ligand

#ATP-binding
#Magnesium
#Manganese
#Metal-binding
#Nucleotide-binding

More keywords

Enzyme and pathway databases

Reactome

R-CFA-70171 Glycolysis
R-CFA-70268 Pyruvate metabolism

UniPathway

UPA00109UER00188

ENZYME

Search...

Names & Taxonomy

Protein names

Recommended name

Pyruvate kinase PKLR

EC number

EC:2.7.1.40 (UniProtKB | ENZYME | Rhea) By Similarity

This website requires cookies, and the limited processing of your personal data in order to function. By using the site you are agreeing to this as outlined in our **Privacy Notice**.

Gene names[i]

Name
PKLR

Organism names

Taxonomic identifier[i]
9615 (NCBI)

Organism[i]
Canis lupus familiaris (Dog) (Canis familiaris)

Strain
Boxer

Taxonomic lineage[i]
Eukaryota > Metazoa > Chordata > Craniata > Vertebrata > Euteleostomi > Mammalia > Eutheria > Laurasiatheria > Carnivora > Caniformia > Canidae > Canis

Accessions[i]

Primary accession
Q29536

Secondary accessions
F1P923
F1PXP8
F1PY57

Proteomes[i]

Identifier
UP000805418

Component[i]
Chromosome 7

Organism-specific databases

VEuPathDB
HostDB:ENSCAFG00845012059

VGNC
VGNC:44609 PKLR

Phenotypes & Variants[i]

This website requires cookies, and the limited processing of your personal data in order to function. By using the site you are agreeing to this as outlined in our **Privacy Notice**.

Defects in PKLR are a cause of inherited hemolytic anemia, an autosomal recessive disease of the Basenji dog that closely resembles human pyruvate kinase deficiency 1 Publication

Expression[i]

Gene expression databases

Bgee

ENSCAFG00000016959 Expressed in liver and 31 other cell types or tissues

Interaction[i]

Subunit[i]

Homotetramer. By Similarity

Protein-protein interaction databases

FunCoup

Q29536 295 interactors

STRING

9615.ENSCAFP00000061352

Structure[i]

3D structure databases

AlphaFoldDB

Q29536

ModBase

Search...

SMR

Q29536

This website requires cookies, and the limited processing of your personal data in order to function. By using the site you are agreeing to this as outlined in our **Privacy Notice**.

Sequence similarities[i]

Belongs to the **pyruvate kinase family**. Curated

Family and domain databases

- View the Phylogenomic databases for this entry within the **Homologs section**.
- View all family and domain features for this entry's canonical sequence in the **UniParc Feature Viewer**.

CDD
cd00288 Pyruvate_Kinase 1 hit

FunFam
2.40.33.10:FF:000001 Pyruvate kinase 1 hit
3.20.20.60:FF:000025 Pyruvate kinase 1 hit
3.40.1380.20:FF:000001 Pyruvate kinase 1 hit

Gene3D
3.20.20.60 Phosphoenolpyruvate-binding domains 1 hit
2.40.33.10 PK beta-barrel domain-like 1 hit
3.40.1380.20 Pyruvate kinase, C-terminal domain 1 hit

InterPro
View protein in InterPro
IPR001697 Pyr_Knase
IPR015813 Pyrv/PenolPyrv_kinase-like_dom
IPR040442 Pyrv_kinase-like_dom_sf
IPR011037 Pyrv_Knase-like_insert_dom_sf
More InterPro links

NCBIfam
NF004491 PRK05826.1 1 hit
NF004978 PRK06354.1 1 hit
TIGR01064 pyruv_kin 1 hit

PANTHER
PTHR11817 PYRUVATE KINASE 1 hit

PRINTS
PR01050 PYRUVTKNASE

PROSITE
View protein in PROSITE
PS00110 PYRUVATE_KINASE 1 hit

Pfam
View protein in Pfam
PF00224 PK 1 hit
PF02887 PK_C 1 hit

SUPFAM
SSF51621 Phosphoenolpyruvate/pyruvate domain 1 hit
SSF50800 PK beta-barrel domain-like 1 hit
SSF52935 PK C-terminal domain-like 1 hit

MobiDB
Search…

Sequence & Isoform[i]

Align isoforms (2) **Add isoforms**

Sequence status[i]
Complete

This website requires cookies, and the limited processing of your personal data in order to function. By using the site you are agreeing to this as outlined in our **Privacy Notice**.

Q29536-1

This isoform has been chosen as the **canonical** sequence. All positional information in this entry refers to it. This is also the sequence that appears in the downloadable versions of the entry.

Name
R-type
Synonyms
PKR

See also
sequence in **UniParc** or sequence clusters in **UniRef**

Length
574

Mass (Da)
62,048

Last updated
2012-03-21 v2

MD5 Checksum[i]
C3E0224B3265F942A70B5C53BC1B81E6

MSSQENIQPQELWSRISKSQRDLAKSILIGAPGGPAGYLRRASVAQLTLELGTAFFQQQQLSAAMADTFLEHL
CLLDIDSEPVAARSTSIIATIGPASHSVERLKEMIKAGMNIARLNFSHGSHEYHAQSIANIREAVESFATSPLGYR
PVAIALDTKGPEIRTGVLKGGPETEVELVKGSWVLVTVDPAFRTLGDAHTVWVDYPNIVKVVPVGGRIFIDDGLI
SLQVKKIDRKGLETQVENGGLLGSRKGVNLPGAEVDLPGLSEQDAQDLRFGVEHNVDIVFASFVRKASDVAAI
RAALGPEGRTIKIISKIENHEGVKKFDEILEVSDGIMVARGDLGIEIPAEKVFLAQKMMIGRCNLAGKPVVCATQ
MLESMITKPRPTRAETSDVANAVLDGADCIMLSGETAKGKFPVEAVKMQHAIAREAEAAVYHRQLFEELRRAA
PLSRDPTEVTAIGAVEAAFKCCAAAIIVLTKTGRSAQLLSRYRPRAAVIAVTRSAQAARQAHLCRGVFPLLYSEP
PEAIWADDVDRRVQFGIESGKLRGFLRVGDLVIVVTGWRPGSGYTNIMRVLSIS

Q29536-2

Name
L-type
Synonyms
PKL

See also
sequence in **UniParc** or sequence clusters in **UniRef**
Differences from canonical
1-33: MSSQENIQPQELWSRISKSQRDLAKSILIGAPG → ME 1 Publication

Computationally mapped potential isoform sequences[i]

There is 1 potential isoform mapped to this entry

Align Add **View all**

Entry	Entry name	Gene name	Length

This website requires cookies, and the limited processing of your personal data in order to function. By using the site you are agreeing to this as outlined in our **Privacy Notice**.

Sequence caution[i]

The sequence **DN338485** differs from that shown. Reason: Miscellaneous discrepancy Derived from EST data. Curated

Features

Showing features for alternative sequence[i], sequence conflict[i].

±	TYPE: All	ID	POSITION(S)	DESCRIPTION	
+	Alternative sequence	VSP_042444	1-33	in isoform **L-type** 1 Publication	Tools Add
+	Sequence conflict		423	in Ref. 2; AAB31627 Curated	
+	Sequence conflict		458-459	in Ref. 2; AAB31627 Curated	

Keywords[i]

Coding sequence diversity

#Alternative splicing

Technical term

#Reference proteome

Sequence databases

RefSeq

NP_001242947.1 **NM_001256018.1**
NP_001243191.1 **NM_001256262.2** [**Q29536-1**]

NUCLEOTIDE SEQUENCE	PROTEIN SEQUENCE	MOLECULE TYPE	STATUS
AAEX03005338 **EMBL** · **GenBank** · **DDBJ**	-	Genomic DNA	No translation available.
AH004600	AAB31627.2		

This website requires cookies, and the limited processing of your personal data in order to function. By using the site you are agreeing to this as outlined in our **Privacy Notice**.

NUCLEOTIDE SEQUENCE	PROTEIN SEQUENCE	MOLECULE TYPE	STATUS
DN338485 EMBL · GenBank · DDBJ	-	mRNA	No translation available.

Genome annotation databases

Ensembl

ENSCAFT00000026853 ENSCAFP00000024956 ENSCAFG00000016959 [Q29536-2]
ENSCAFT00000035475 ENSCAFP00000030738 ENSCAFG00000016959 [Q29536-1]
ENSCAFT00030047792 ENSCAFP00030041802 ENSCAFG00030025851 [Q29536-1]
ENSCAFT00030047805 ENSCAFP00030041815 ENSCAFG00030025851 [Q29536-2]
ENSCAFT00040001436 ENSCAFP00040001228 ENSCAFG00040000772 [Q29536-1]

More Ensembl links

GeneID

490425

KEGG

cfa:490425

Similar Proteins[i]

UniRef clusters[i]

Homologs[i]

Orthologs & paralogs[i]

Phylogenomic databases

This website requires cookies, and the limited processing of your personal data in order to function. By using the site you are agreeing to this as outlined in our **Privacy Notice**.

OrthoDB
108365at2759

eggNOG
KOG2323 Eukaryota

Disclaimer

Any medical or genetic information present in this entry is provided for research, educational and informational purposes only. It is not in any way intended to be used as a substitute for professional medical advice, diagnosis, treatment or care. Our staff consists of biologists and biochemists that are not trained to give medical advice.

© 2002 – 2026 UniProt consortium

This website requires cookies, and the limited processing of your personal data in order to function. By using the site you are agreeing to this as outlined in our **Privacy Notice**.